M&AE 4070/5070 – Dynamics of Flight Vehicles

Case Study 1

Performances of a high-altitude, long-endurance drone

Introduction. In 1986, Boeing developed the Condor, a high-altitude, long-endurance plane. This plane is unmanned, with a span higher than a Boeing 747, and is able to fly several days at twice the altitude of commercial planes. The Condor project stayed classified for a long time, with very little information leaking to the (specialized) press. In this case study, we are going to try to recover the mass, endurance, and ceilings of this plane from the incomplete set of data available at the time.



Available data

- Wing span: $b = 60$ m
- Aspect ratio: $\mathbf{AR} = 36$. Also assume that the effective aspect ratio is equal $\mathbf{AR}$, that is, $e\mathbf{AR} = \mathbf{AR} = 36$.
- Maximum Lift-to-Drag ratio = 41
- Mean $V_{\text{cruise}} = 225$ km/h
- $h_{\text{cruise}} = 45,000$ ft
- The Condor is equipped with two turbo-charged **piston** engines, which mean that *they can provide the same power from ground level to absolute ceiling.* The power delivered from *each* engine is $P_E = 130$ kW, and the propeller has an efficiency of $\eta_P = 0.85$. This means that the actual power generated by the propeller is $\eta_P P_E$.
- The specific consumption of each engine is $C_{s_P} = 0.25$ kg/kW.h (Note the units: this is the thrust-specific fuel consumption for a **propeller** engine, which is different from the jet engines thrust-specific fuel consumption C_t you may have seen in MAE 3050.)

Questions

1. What is the density at the cruising altitude of the Condor?

The flight altitude is above the 11 km cutoff separating linear temperature decrease from constant value in standard T profile

1. Find density at 11 km using linear decrease formula
2. Use ρ_{11km} to get ρ @ $h_{condor} = 13,716$ m

- $\rho_{11km} = \rho_{SSL}\left(1 + \frac{a h_{11km}}{T_{SSL}}\right)^{-\frac{g_0}{aR}-1}$

$g_0 = 9.81\ m/s^2$, $R = 287\ J/kg.K$, $a = -0.0065\ K/m$, $\rho_{SSL} = 1.225\ kg/m^3$, $T_{SSL} = 288.15K$

$$\rho_{11km} = 1.225\left(1 - \frac{0.0065 \times 11{,}000}{288.15}\right)^{-\frac{9.81}{-0.0065 \times 287}-1}$$

$$= 0.364\ kg/m^3$$

- $\rho_{h_{condor}} = \rho_{11km}\, e^{-\frac{g_0}{RT_{11km}}(h_{condor} - h_{11km})}$

$T_{above\ 11km} = 216.67K$

$$= \boxed{0.237\ kg/m^3}$$

2. From the provided data, what is the reference surface area S for this plane?

Aspect ratio: $AR = \frac{b^2}{S}$

with b: wing span = 60 m

S: surface area of the wing

AR: aspect ratio = 36

$$\Rightarrow \quad S = \frac{b^2}{AR} = \frac{60^2}{36} = \boxed{100\ m^2}$$

3. Derive the expression for the available flight time (or max endurance) of the Condor for **a Breguet** cruise control law (altitude and angle of attack constant) and check that it matches the corresponding formula provided in lecture notes.

Endurance, propeller engine : $E = \int_{t_i=0}^{t_f} dt = -\int_{m_i}^{m_f} \frac{dm}{C}$

$C \equiv$ fuel consumption rate $= \frac{C_{sp}}{\eta_p} \frac{1}{2} \rho S V^3 C_x$ } accounts for both engines already since it corresponds to the thrust needed to compensate for the drag during the cruise.

L/D $E = -\int_{m_i}^{m_f} \frac{\eta_p}{C_{sp} \frac{1}{2} \rho S V^3 C_x} dm$

To go further, we need to identify what is constant inside the integral.

We assume the Breguet control law : keeping α and h (and therefore density) constant throughout flight. that means V varies and it should therefore be replaced by a function of C_z

$$L = W = mg = \frac{1}{2} \rho S V^2 C_z \Rightarrow \quad V^3 = \left(\frac{mg}{\frac{1}{2} \rho S C_z} \right)^{3/2}$$

Endurance becomes:

$$E = -\int_{m_i}^{m_f} \frac{\eta_p}{C_{sp}} \frac{1}{\frac{1}{2} \rho S C_x} \left(\frac{\frac{1}{2} \rho S C_z}{mg} \right)^{3/2} dm$$

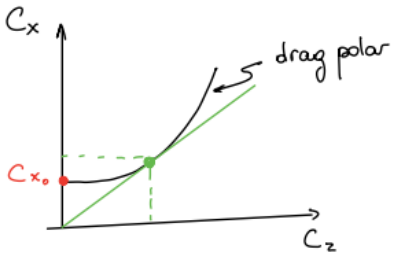
$$E = -\int_{m_i}^{m_f} \underbrace{\frac{\eta_p}{C_{sp}} \frac{\sqrt{\frac{1}{2} \rho S}}{g^{3/2}}}_{\text{constant}} \underbrace{\frac{C_z^{3/2}}{C_x}}_{\text{constant if } \alpha \text{ is constant}} \frac{dm}{m^{3/2}}$$

$$E = -\frac{\eta_p}{C_{sp}} \frac{\sqrt{\frac{1}{2} \rho S}}{g^{3/2}} \frac{C_z^{3/2}}{C_x} \underbrace{\int_{m_i}^{m_f} \frac{dm}{m^{3/2}}}_{= \left[-\frac{2}{\sqrt{m}} \right]_{m_i}^{m_f} = -2 \left[\frac{1}{\sqrt{m_f}} - \frac{1}{\sqrt{m_i}} \right]}$$

Finally

$$\boxed{E = \frac{\eta_p}{C_{sp}} \frac{\sqrt{2\rho}}{g^{3/2}} \sqrt{\frac{S}{m_i}} \frac{C_z^{3/2}}{C_x} \left(\frac{1}{\sqrt{1 - w_m}} - 1 \right)}$$

4. What is the drag coefficient at zero lift C_{x_0} for this plane? Is this value realistic?



Only info related to drag polar in data provided is the MAXIMUM lift to drag ration:

$$f_{max} = \left.\frac{C_z}{C_x}\right|_{max} = 41$$

which corresponds to the slope of the line going through the origin and tangent to the drag polar

$\Rightarrow$ use f_{max} and the drag polar equation to extract value for C_{x_0}.

* $\frac{C_z}{C_x}$ is max when $\frac{C_x}{C_z}$ is minimized.

* $C_x = C_{x_0} + \frac{1}{\pi AR} C_z^2$

* Extremum of $\frac{C_x}{C_z}$ is obtained when $\frac{d}{dC_z}\left(\frac{C_x}{C_z}\right) = 0$

$$\Rightarrow \frac{d}{dC_z}\left(\frac{C_{x_0} + \frac{C_z^2}{\pi AR}}{C_z}\right) = \frac{d}{dC_z}\left(\frac{C_{x_0}}{C_z} + \frac{C_z}{\pi AR}\right) = 0$$

$$\Rightarrow -C_{x_0}\frac{1}{C_z^2} + \frac{1}{\pi AR} = 0$$

$$\Rightarrow C_z\Big|_{f_{max}} = \sqrt{C_{x_0}\pi AR}$$

* $$f_{max} = 41 = \frac{\sqrt{C_{x_0}\pi AR}}{C_{x_0} + \frac{C_z^2}{\pi AR}} = \frac{\sqrt{C_{x_0}\pi AR}}{C_{x_0} + \frac{\pi AR C_{x_0}}{\pi AR}} = \frac{\sqrt{C_{x_0}\pi AR}}{2C_{x_0}}$$

$$\Rightarrow \sqrt{C_{x_0}} = \frac{\sqrt{\pi AR}}{2 f_{max}} = \frac{\sqrt{\pi \times 36}}{2 \times 41} = 0.130$$

$$\Rightarrow \boxed{C_{x_0} = 0.0168}$$

Order of magnitude is correct. (expects $\sim 10^{-2}$)

5. Evaluate the cruise lift and drag coefficients for the Condor, accounting for the fact that:
   - we want to maximize endurance,
   - realistically speaking, the maximum lift coefficient for such a plane is at most 1.85,
   - the plane flies with a "stall margin" such that the cruise velocity V is 20% higher than the stall velocity V_S:

$$V = 1.2V_S \tag{1}$$

* The aircraft's mission is surveillance $\Rightarrow$ it will fly in such conditions as to maximize endurance. Since it is a propeller driven aircraft the proper function of $\frac{C_z}{C_x}$ that needs to be maximized is $\frac{C_z^{3/2}}{C_x}$

* $\frac{C_z^{3/2}}{C_x}$ is maximized when $C_z = \sqrt{3\pi AR C_{x_0}}$

* The optimal lift coefficient is therefore

$$C_z = \sqrt{3\pi \times 36 \times 0.0168} = \underline{2.39}$$

This optimal value is unrealistic, since stall happens at $C_{z_{max}} = 1.85$

$$C_{z_{max}} \Longleftrightarrow V_{stall}$$

what we are looking for, value should be smaller than $C_{z\,max}$ $\longrightarrow C_z \Longleftrightarrow 1.2\,V_{stall}$ (since $L = W = \frac{1}{2}\rho V^2 S C_z$, V will need to be higher than V_{stall} since $C_z < C_{z_{max}}$)

and

$$C_z = \frac{W}{\frac{1}{2}\rho S V^2}$$

$$\Rightarrow \frac{C_{z_{max}}}{C_z} = \frac{(1.2\,V_{stall})^2}{V_{stall}^2}$$

$$\Rightarrow C_z = \frac{C_{z_{max}}}{1.44} = \frac{1.85}{1.44} = \boxed{1.28 = C_z}$$

The corresponding drag coefficient is obtained from the drag polar:

$$C_x = C_{x_0} + \frac{C_z^2}{\pi AR} = 0.018 + \frac{1.25^2}{\pi \times 37} = \boxed{0.031 = C_x}$$

6. Evaluate the mean m/S ratio and the mass at take-off. The mean m/S ratio should be computed from the mean cruise velocity. The *initial* m/S can then be evaluated using the mass coefficients v_m and u_m:

   - v_m is the ratio between empty mass and initial mass, and is typically equal to 0.5.
   - u_m is the ratio between the mass of the payload and the initial mass, which is set to 2% (*i.e.* $u_m = 0.02$) in this case. $u_m = 0$ corresponds to maximum flight time without any payload, and increasing u_m leads to performances associated with heavier payload.

   Note that the mean mass is defined as the average between initial and final weight:

$$m_{\text{mean}} = \frac{m_i + m_f}{2} \tag{2}$$

- Cruise conditions — equilibrium state

$V_{mean} = 225 \text{ km/h} = \frac{225 \times 1000}{3600} = 62.5 \text{ m/s}$

Lift = Weight or $\frac{1}{2}\rho S V^2 C_z = mg$

So $\frac{m_{mean}}{S} = \frac{1}{2}\frac{\rho}{g} C_z V_{mean}^2 = \frac{1}{2}\frac{0.237}{9.81} \times 1.28 \times 62.5^2 = \boxed{60.4 \text{ kg/m}^2}$

- At take-off

$$\frac{m_{mean}}{S} = \frac{m_i + m_f}{2S} = \frac{m_i\left(1 + v_m + u_m\right)}{2S}$$

$$\frac{m_i}{S} = \frac{2}{1 + v_m + u_m}\left(\frac{m_{mean}}{S}\right) = \frac{2}{1 + 0.5 + 0.02} \times 60.4 = \boxed{79.5 \text{ kg/m}^2}$$

7. How long can the Condor actually stay in the air, that is, what is its Breguet endurance with the parameters provided/evaluated so far? Make sure your answer is reasonable.

$$E = \frac{0.85 \times 1000 \times 3600}{0.25} \frac{\sqrt{2 \times 0.237}}{9.81^{3/2}} \sqrt{\frac{1}{79.5}} \frac{1.28^{3/2}}{0.031}\left(\frac{1}{\sqrt{0.5 + 0.02}} - 1\right)$$

$E = 549\,548 \text{ s}$

Converting to hours:

$\boxed{E = 154 \text{ hours} = 6.4 \text{ days}}$

Those values are those found in previous question to avoid stall

$\Rightarrow$ very close to actual value that got available later on

8. The absolute ceiling (also called propulsion ceiling) of an aircraft is the the highest altitude at which an aircraft can fly with its engines pushed to their maximum. At the absolute ceiling, the maximum power/thrust available from the engines are used to maintain speed and altitude. Evaluate the absolute ceiling of the Condor, based on the mean mass computed above, and the maximum realistic angle of attack (or maximum realistic C_z), also determined above. At which velocity will the aircraft fly when at its absolute ceiling?

Propulsion or absolute ceiling is obtained by finding the minimum density at which thrust can still compensate drag.

$$\text{Thrust} = \text{Drag}$$

$$\underbrace{P_A}_{\text{(max) power available from both engines}} = \text{Thrust}\, V = \frac{1}{2}\rho S V^2 C_x V \qquad (1)$$

$$P_A = \eta_p (2P_E)$$

Aircraft still has to fly: $mg = \frac{1}{2}\rho S V^2 C_z \qquad (2)$

$$V^2 = \frac{mg}{\frac{1}{2}\rho S C_z} \quad \text{or} \quad V^3 = \left(\frac{mg}{\frac{1}{2}\rho S C_z}\right)^{3/2}$$

(1) becomes $2\eta P_E = \frac{1}{2}\rho S C_x \left(\frac{mg}{\frac{1}{2}\rho S C_z}\right)\sqrt{\frac{2mg}{\rho S C_z}}$

$$\Rightarrow \quad (2\eta P_E)^2 = C_x^2 \frac{(mg)^2}{C_z^2}\frac{2mg}{\rho S C_z}$$

$$\Rightarrow \quad \rho = \frac{1}{2}\frac{1}{(\eta P_E)^2}\frac{C_x^2}{C_z^3}\frac{(mg)^3}{S}$$

so $\rho = \frac{1}{2}\frac{1}{(0.85 \times 130 \times 10^3)^2}\frac{\left(0.0168 + \frac{1}{\pi \times 36}(1.85)^2\right)^2}{1.85^3}\frac{\overbrace{(60.4 \times 100}^{m_{mean}} \times 9.81)^3}{100}$

$\rho = 0.0298 \text{ kg/m}^3$

max realistic C_z

Density is below density at 11 km : need to use exponential formula

$$\rho = \rho_{11\,km}\exp\left(-\frac{g}{RT_{11km}}(h - h_{11km})\right)$$

$$h = -\frac{RT_{11km}}{g}\ln\left(\frac{\rho}{\rho_{11km}}\right) + h_{11km}$$

$$h = -\frac{287 \times 216.67}{9.81}\ln\left(\frac{0.0298}{0.3639}\right) + 11{,}000$$

$$\boxed{h = 26\,862 \text{ m}}$$

The velocity of the Condor at this absolute ceiling is obtained from

$$V = \left(\frac{mg}{\frac{1}{2}\rho S C_z}\right)^{1/2} = \left(\frac{60.4 \times 9.81}{0.5 \times 0.0298 \times 1.85}\right)^{1/2} = 146.6 \text{ m/s} = 527.8 \text{ km/h}$$

much higher than V_{cruise}!

9. The sustentation ceiling is defined as the maximal theoretical altitude at which a plane can fly level, *independently from its motorization*. Defined for an aircraft's maximum lift coefficient C_z and maximum Mach number, it is usually higher than the absolute ceiling. Check that it is indeed the case for the Condor, given the data that we now have. For the Condor, $M_{\max} = 0.7$.

In steady flight, assuming C_z and Mach are maximum, we have:

$$m_{mean}\, g = \frac{1}{2}\rho V_{max}^2 S C_{z_{max}} \quad (1)$$

Mach number $M = \frac{V}{a}$, where $a = \sqrt{\gamma RT} = \sqrt{\gamma \frac{P}{\rho}}$ is the speed of sound and γ is the ratio of specific heats ($= 1.4$ for standard air)

so $V_{max}^2 = M_{max}^2 \gamma \frac{P}{\rho}$ and (1) simplifies into:

$P_{ceiling}$ → unknown!

$$m_{mean}\, g = \frac{1}{2}\rho M_{max}^2 \gamma \frac{P}{\rho} S C_{z_{max}}$$

$$P_{ceiling} = \frac{2 m_{mean} g}{\gamma M_{max}^2 S C_{z_{max}}} = \frac{2 \times 60.4 \times 100 \times 9.81}{1.4 \times 0.7^2 \times 100 \times 1.85}$$

$$P_{ceiling} = 933.8 \text{ Pa}$$

We can convert p into a corresponding altitude using the standard atmosphere assumption:

$$P_{11km} = P_{SSL}\left(1 + \frac{a h_{11km}}{T_{SSL}}\right)^{-g/aR} = 22\,628 \text{ Pa}$$

$$P = P_{11km} \exp\left(-\frac{g}{RT_{11km}}(h - h_{11km})\right)$$

so $h_{ceiling} = \ln\left(\frac{P_{ceiling}}{P_{11km}}\right)\left(-\frac{RT_{11km}}{g}\right) + h_{11km} = 31{,}206 \text{ m} = \boxed{31 \text{ km}}$

As expected, sustentation ceiling > propulsion/absolute ceiling